ClipSAT : ACT International Subject Test: Mathematics Level 2 [ANSWER KEY]

Unit 7.01: Combinatorics, Binomial Probability & Regression

Q1. The graph of the function fx = x - 1^2 - 4 is shown in the standard x, y coordinate plane below. What is the range of f ?

[x] A: [-4, infty

[ ] B: [1, infty

[ ] C: -infty, -4]

[ ] D: -infty, infty

Solution: 1. **Identify the vertex and direction of opening**: The function fx = x - 1^2 - 4 is written in vertex form, fx = ax - h^2 + k , where a = 1 , h = 1 , and k = -4 . The vertex of the parabola is at 1, -4. 2. **Determine the minimum output value**: Since the leading coefficient a = 1 > 0 , the parabola opens upward, meaning that the vertex represents the absolute minimum point on the graph. The minimum value attained by fx is -4. 3. **Determine the range**: The range represents all possible output values y-values of the function. Because the function takes on its minimum value of -4 at x = 1 and increases without bound as x moves away from 1, the range is all real numbers greater than or equal to -4, which is written in interval notation as [-4, infty. **Common Misconceptions:** - Choice B, [1, infty, incorrectly uses the x-coordinate of the vertex instead of the y-coordinate. - Choice C, -infty, -4], incorrectly assumes the parabola opens downward. - Choice D, -infty, infty, confuses the range the set of y-values with the domain the set of all real x-values.

Q2. For the complex number i = sqrt{-1}, which of the following is equivalent to the expression 5 - 3i - 2 - 7i?

[x] A: 3 + 4i

[ ] B: 3 - 10i

[ ] C: 7 + 4i

[ ] D: 3 - 4i

Solution: To subtract complex numbers, distribute the negative sign across the second term and combine like terms real parts with real parts, and imaginary parts with imaginary parts: 5 - 3i - 2 - 7i = 5 - 3i - 2 + 7i Group the real and imaginary components: - Real part: 5 - 2 = 3 - Imaginary part: -3i + 7i = 4i Combining these gives 3 + 4i. Common misconception checks: - Choice B 3 - 10i occurs if the negative sign is not distributed to the imaginary term, mistakenly calculating -3i - 7i = -10i. - Choice C 7 + 4i results from accidentally adding the real parts 5 + 2 = 7 while subtracting the imaginary parts. - Choice D 3 - 4i results from a sign error when computing -3 - -7.

Q3. What is the value of lim_{x to 3} frac{x^2 - 2x - 3}{x - 3} ?

[x] A: 4

[ ] B: 0

[ ] C: 6

[ ] D: The limit does not exist

Solution: Direct substitution of x = 3 into the expression gives frac{3^2 - 23 - 3}{3 - 3} = frac{0}{0} , which is an indeterminate form. Factor the numerator: x^2 - 2x - 3 = x - 3x + 1 For all x neq 3 , simplify the rational expression by canceling the common factor x - 3 : frac{x - 3x + 1}{x - 3} = x + 1 Now evaluate the limit by direct substitution: lim_{x to 3} x + 1 = 3 + 1 = 4 Common misconceptions include concluding that the limit is 0 because the numerator evaluates to 0 , or incorrectly believing that the limit does not exist simply because the denominator evaluates to 0 at x = 3 .

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Q4. Given the vectors mathbf{u} = langle 3, -4 rangle and mathbf{v} = langle -2, 5 rangle , what is the vector 2mathbf{u} + mathbf{v} ?

[x] A: langle 4, -3 rangle

[ ] B: langle 1, 1 rangle

[ ] C: langle 4, 1 rangle

[ ] D: langle 8, -13 rangle

Solution: To find 2mathbf{u} + mathbf{v} , first perform the scalar multiplication on mathbf{u} : 2mathbf{u} = 2langle 3, -4 rangle = langle 23, 2-4 rangle = langle 6, -8 rangle Next, add the corresponding components of 2mathbf{u} and mathbf{v} : 2mathbf{u} + mathbf{v} = langle 6 + -2, -8 + 5 rangle = langle 4, -3 rangle - langle 1, 1 rangle is incorrect because it represents mathbf{u} + mathbf{v} , omitting the scalar factor of 2. - langle 4, 1 rangle is incorrect because the scalar 2 was only distributed to the horizontal component. - langle 8, -13 rangle is incorrect because it calculates 2mathbf{u} - mathbf{v} instead of 2mathbf{u} + mathbf{v} .

Q5. The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?

[ ] A: 131

[x] B: 138

[ ] C: 140

[ ] D: 145

Solution: To find the 20th term of the arithmetic sequence, first identify the initial term and the common difference: 1. The first term is a_1 = 5 . 2. The common difference d is found by subtracting consecutive terms: d = 12 - 5 = 7 . The general formula for the n th term of an arithmetic sequence is: a_n = a_1 + n - 1d Substitute n = 20 , a_1 = 5 , and d = 7 : a_{20} = 5 + 20 - 17 = 5 + 197 = 5 + 133 = 138 Common misconceptions: - Using n instead of n - 1 yields 5 + 207 = 145 . - Calculating 20 times 7 = 140 overlooks the starting value a_1 = 5 . - Using n - 2 steps yields 5 + 187 = 131 .

Q6. A point has polar coordinates r, theta = left6, frac{2pi}{3}right. What are the Cartesian coordinates x, y of this point?

[x] A: -3, 3sqrt{3}

[ ] B: -3sqrt{3}, 3

[ ] C: 3, -3sqrt{3}

[ ] D: 3sqrt{3}, -3

Solution: To convert polar coordinates r, theta to Cartesian coordinates x, y, use the conversion formulas: x = r costheta quad text{and} quad y = r sintheta Given r = 6 and theta = frac{2pi}{3}: x = 6 cosleftfrac{2pi}{3}right = 6 left-frac{1}{2}right = -3 y = 6 sinleftfrac{2pi}{3}right = 6 leftfrac{sqrt{3}}{2}right = 3sqrt{3} Thus, the Cartesian coordinates are -3, 3sqrt{3}. Common errors: - Interchanging x and y using x = rsintheta and y = rcostheta yields 3sqrt{3}, -3. - Reversing the quadrant signs or applying frac{pi}{6} reference angle formulas erroneously produces -3sqrt{3}, 3.

Q7. The polynomial function Px = 2x^3 + ax^2 - 5x + b has x - 1 as a factor. When Px is divided by x + 2 , the remainder is -18 . What is the remainder when Px is divided by x - 3 ?

[x] A: 2

[ ] B: 8

[ ] C: 18

[ ] D: -30

Solution: According to the Factor Theorem and the Remainder Theorem, if a polynomial Px is divided by x - c , the remainder is Pc . If x - c is a factor, then Pc = 0 . Step 1: Set up the first equation using the factor x - 1 : P1 = 21^3 + a1^2 - 51 + b = 0 2 + a - 5 + b = 0 implies a + b = 3 Step 2: Set up the second equation using the divisor x + 2 = x - -2 with remainder -18 : P-2 = 2-2^3 + a-2^2 - 5-2 + b = -18 2-8 + 4a + 10 + b = -18 -16 + 10 + 4a + b = -18 implies 4a + b = -12 Step 3: Solve the system of linear equations for a and b : Subtracting the first equation from the second: 4a + b - a + b = -12 - 3 3a = -15 implies a = -5 Substitute a = -5 into a + b = 3 : -5 + b = 3 implies b = 8 Thus, the polynomial is Px = 2x^3 - 5x^2 - 5x + 8 . Step 4: Find the remainder when Px is divided by x - 3 by evaluating P3 : P3 = 23^3 - 53^2 - 53 + 8 = 227 - 59 - 15 + 8 = 54 - 45 - 15 + 8 = 2 Common Pitfalls: -

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Q8. For what real value of k is the complex expression frac{k + 8i}{3 - i} + frac{2 - ki}{3 + i} a real number?

[x] A: 11

[ ] B: 1

[ ] C: 6

[ ] D: -11

Solution: To determine when the sum of the two fractions is a real number, simplify the expression by combining terms over a common denominator or multiplying by conjugates. The common denominator of the two fractions is: 3 - i3 + i = 3^2 - i^2 = 9 - -1 = 10 Rewrite each term over this denominator: 1. For the first term: frac{k + 8i3 + i}{3 - i3 + i} = frac{3k + ki + 24i + 8i^2}{10} = frac{3k - 8 + k + 24i}{10} 2. For the second term: frac{2 - ki3 - i}{3 + i3 - i} = frac{6 - 2i - 3ki + ki^2}{10} = frac{6 - k + -3k - 2i}{10} Now, sum the real and imaginary components: text{Real part} = frac{3k - 8 + 6 - k}{10} = frac{2k - 2}{10} text{Imaginary part} = frac{k + 24 + -3k - 2}{10} = frac{-2k + 22}{10} For the resulting complex number to be a real number, its imaginary part must equal 0 : frac{-2k + 22}{10} = 0 implies -2k + 22 = 0 implies 2k = 22 implies k = 11 *Common Misconceptions:* - Choosing **1** results from setting the real part equal to 0 2k - 2 = 0 , which would make the expression purely imaginary rather than real. - Choosing **6** results from equating the real and imaginary parts 2k - 2 = -2k + 22 . - Choosing **-11** results from an algebraic sign error when solving -2k = -22 .